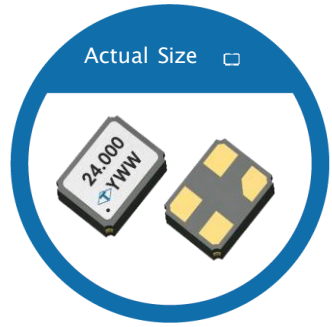
TAITIEN
Our Performance, Your Reputation

O3 Type
1.6 x 1.2 mm SMD Crystal Oscillator



RoHS Compliant

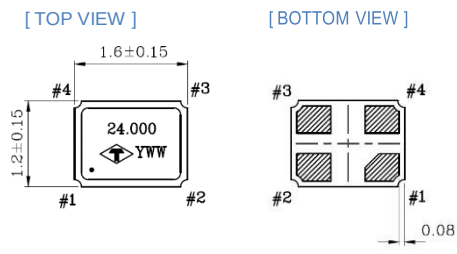
FEATURE

- Typical 1.6 x 1.2 x 0.8 mm ceramic SMD package
- Low Power Supply Voltage: 3.3, 2.5, and 1.8V supply options
- Clock Output: CMOS
- Tight symmetry (45 to 55%) available.
- Tri-state enable / disable mode.
- Pb-free/RoHS compliant

TYPICAL APPLICATION

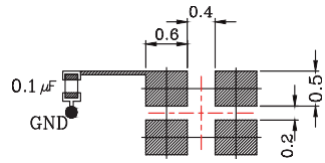
- WLAN
- Mobile Phone
- DSC, Set-top Box, HDTV

DIMENSION (mm)



Pin#	Function
1	TRI-STATE
2	GND
3	Output
4	VDD

SOLDER PAD LAYOUT (mm)



To ensure optimal oscillator performance,
place a by-pass capacitor of 0.1μF
as close to the part as possible between Vdd and GND pads.

ELECTRICAL SPECIFICATION

Parameter		CMOS						Unit
		3.3V		2.5V		1.8V		
		Min.	Max.	Min	Max.	Min	Max.	
Supply Voltage Variation (V_{DD})		2.97	3.63	2.25	2.75	1.71	1.89	V
Frequency Range		3	60	3	60	3	60	MHz
Standard Frequency		10,12,12.288,19.2,20,24,25,40,50						MHz
Supply Current		-	15	-	15	-	15	mA
Standby Current		-	10	-	10	-	10	uA
Duty cycle		45	55	45	55	45	55	%
Output Level (CMOS)	Output High	0.9 x V_{DD}	-	0.9 x V_{DD}	-	0.9 x V_{DD}	-	V
	Output Low	-	0.1 x V_{DD}	-	0.1 x V_{DD}	-	0.1 x V_{DD}	V
Transition Time (20% - 80%)	Rise Time	-	10	-	10	-	10	nSec
	Fall Time	-	10	-	10	-	10	nSec
Startup Time		-	2	-	2	-	2	mSec
Tri-State mode	Enable	0.7 x V_{DD}	-	0.7 x V_{DD}	-	0.7 x V_{DD}	-	V
(Input to Pin 1)	Disable	-	0.3 x V_{DD}	-	0.3 x V_{DD}	-	0.3 x V_{DD}	V
Period Jitter(Pk-Pk)		-	40	-	40	-	40	ps
RMS Phase Jitter (Integrated 12 KHz~20 MHz)		-	1	-	1	-	1	ps
Output Driver Capability (CL)		-	15	-	15	-	15	pF
Aging(@25°C 1st year)		-	±3	-	±3	-	±3	ppm
Operating temp. range		-40	105	-40	105	-40	105	°C
Storage temp		-55	125	-55	125	-55	125	°C

FREQ. STABILITY vs. TEMP. RANGE

ppm / Temp. (°C)	±20	±25	±50	±100
-20~+70	O	O	O	O
-30~+85	△	O	O	O
-40~+85	△	O	O	O
-40~+105	X	X	O	O

* ○: Available △:Conditional X: Not available

* Inclusive of calibration @ 25 °C, operating temperature range, input voltage variation, load variation, aging (1st year), shock, and vibration

Note: notall combination of options are available. Other specifications may be available uponrequest.